Balaji K

+91 81488 39209 | balaji.k2022b@vitstudent.ac.in | linkedin.com/in/balaji-k | github.com/balaji2309

Professional Summary

Software Engineering student specializing in GenAI, Agentic AI systems, and Data Engineering, with interdisciplinary experience spanning AI-driven software and industrial IoT systems. Experienced in building scalable data pipelines, RAG-based applications, and multi-agent AI systems for real-world automation. Proven ability to transform manual workflows into efficient, production-ready solutions in Agile environments..

Technical Skills

Languages: Python, C, C++, Java, JavaScript, SQL
AI/ML: GenAI, LLMs, RAG, Agentic AI, PyTorch, XGBoost, NLP
Data Engineering: ETL/ELT, Data Pipelines, Feature Engineering, Pandas, NumPy
Frameworks & Web: FastAPI, Node.js, Streamlit
Tools & Platforms: Git, GitHub, Agile/Scrum, Selenium

Experience

CCPS Data & Research Intern — May 2025 – Present
Vellore Institute of Technology — *Chennai, India*

- **Problem:** High-volume sensor data lacked automated processing, causing delays in multi-temporal analysis.
- **Solution:** Built Python-based data preprocessing pipelines and an ensemble ML framework using MLOps practices.
- **Impact:** Reduced data processing time by 40% and improved pipeline reliability for large-scale datasets.

Project Intern (Automation) — June 2025 – July 2025
Chennai Port Authority — *Chennai, India*

- **Problem:** Manual tracking of port incentives led to inefficiencies and data inconsistencies.
- **Solution:** Automated backend workflows by restructuring data flow and validation logic.
- **Impact:** Decreased manual effort by 60% and improved reporting accuracy and system responsiveness.

AI & Enterprise Innovation Projects

MedAssist: Multi-Agent AI System | *FastAPI, Gemini 1.5 Flash, Agentic AI*

- Built and optimized a **Multi-Agent System (MAS)** to automate medical adherence logging using natural language.
- Developed reusable Python-based frameworks for medical escalation protocols, aligning with enterprise data strategy.
- Built a RAG-based system enabling real-time retrieval of medical protocols for automated decision support.

Multimodal Software Defect Prediction | *PyTorch, NLP, MLOps*

- Developed an AI/ML model to automate bug detection, contributing to all phases of the software development process.
- Performed data transformation and feature engineering on GitHub metadata to support predictive analytics.

Utility Patent: Automated Industrial IoT Fault Detection — 2025

- Invented a LoRaWAN-based system for automated grid monitoring; shortlisted for **VNEST Startup Fund**.

Self-Pruning Neural Network (CIFAR-10) | *PyTorch, Deep Learning*

- Built a custom **PrunableLinear layer** using STE for dynamic weight pruning during training.
- Applied **L1 regularization** to achieve **98.2% sparsity** with minimal accuracy loss.
- Optimized training with cosine scheduling, warmup, and gradient clipping.
- Demonstrated **trade-off**: best at =1e-2 (56.75% acc, 93.7% sparsity).

Education

Vellore Institute of Technology (VIT) — Chennai, India
Integrated M.Tech in Software Engineering ***(CGPA: 7.94 / 10.0)*** — *Expected May 2027*

Certifications

- **Agile Scrum in Practice (Infosys)** – Proficiency in Agile way of working and requirement clarifications.
- **IBM Z Day 2025** – Specialized in Enterprise Computing, Secure AI Integration, and Data Science.
- **Everyday Excel with Macros (University of Colorado) – Intermediate** – Advanced analytics and automated reporting skills.